LINGVESSA

Implementation Plan

Production-oriented bilingual reference implementation / v1.4

Execution plan that converts the synchronized owner-locked Lingvessa specifications into an ordered engineering, content, media, QA and AI delivery backlog with explicit dependencies, gates and exit evidence.

START ENGLISH E-A1-M01 + ČEŠTINA C-A1-M01
ARCHITECTURE MODULAR MONOLITH + WORKERS + POSTGRESQL
QUALITY 32 REFERENCE REGRESSION SCENARIOS + RELEASE GATES
AI OPTIONAL / FREE-LOCAL-FIRST / PAID BUDGET 0
DELIVERY PRIVATE GIT + PORTABLE CONTAINERS + VERSIONED CONTENT/MEDIA

Status: OWNER-LOCKED IMPLEMENTATION BASELINE

Owner lock: 10 September 2026 | Document version date: 09.09.2026 | Replaces Implementation Plan v1.3

Паспорт документа

Поле	Значення
Ідентифікатор	LINGVESSA-IMPL-PLAN
Версія / дата	1.4 / 09.09.2026
Статус	OWNER-LOCKED IMPLEMENTATION BASELINE; owner lock 10.09.2026; P0 authorized.
Продукт	Lingvessa
Стартова реалізація	English E-A1-M01 + Čeština C-A1-M01 на одному shared engine.
Governing baseline	Lingvessa_Formal_Specification v1.7; Lingvessa_Technical_Specification v1.7; Lingvessa_Technical_Design v1.7; Lingvessa_Curriculum_Blueprint_A1 v1.6; Lingvessa_Reference_Module_Design v1.6; Lingvessa_AI_Integration_Specification v1.4.
Execution model	Dependency-gated milestones. Calendar estimates are intentionally not fixed until implementation capacity is known.
Development default	Private Git repository; GitHub Private + Northflank Developer Sandbox + Cloudflare R2 are current development defaults, not production lock-in.
Core invariant	Published lessons, attempts, mastery, Review, progress and standard media remain usable with all AI providers disabled.

Baseline precedence
This plan treats the six currently uploaded/renamed Lingvessa documents as the current project baseline. Where an internal dependency line in an earlier-generated document still names a predecessor version, the later approved document and explicit project decision take precedence. No implementation task may silently change a governing invariant.

Що цей документ додає

Область	Рішення v1.4
Delivery order	Один dependency-driven маршрут від repository skeleton до bilingual reference baseline, потім контрольоване масштабування A1.
Milestones	P0-P13 з Definition of Ready, scope, exit evidence та block conditions.
Parallel work	Content/media authoring може починатися після schema freeze, але bulk publication чекає reference validation.
Quality	Regression, a11y, privacy, security, backup/restore, provider-offline та AI budget gates вбудовані в milestones.
Change control	Contract-changing defect не маскується workaround-ом: створюється ADR/spec revision + migration/test impact.
No fake deadlines	Відсутні вигадані строки. План вимірює готовність доказами, а не календарем.

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1. Призначення, межі та правила виконання

Implementation Plan перетворює вже затверджену продуктову, технічну, curriculum, reference-module та AI документацію Lingvessa у порядок реальної розробки. Це не нова специфікація функцій: він не додає product scope, а визначає, що реалізовувати спочатку, що може йти паралельно, які докази завершення потрібні і коли дозволено переходити до наступного milestone.

- Programming starts with a production-oriented bilingual reference implementation, not a disposable one-folder prototype.
- English and Czech are both real launch paths; Czech must not be postponed into a fake placeholder after English.
- Bulk A1 content does not start before shared contracts are proven by the reference modules.
- AI is added after deterministic core behavior is functioning; AI-off remains a permanent release gate.
- Calendar dates and sprint counts are not normative in v1.4 because team size and implementation throughput are not fixed.
- Each milestone closes only with executable evidence: tests, manifests, QA reports, migrations, release records or approved decisions.

1.1. Не входить у цей implementation baseline

Поза critical path	Правило
Native iOS/Android	Deferred until PWA telemetry/need.
Payments/subscriptions	Separate commercial decision.
Official CEFR certification	Not implied by internal readiness.
PDF/textbook import	Out of v1.
Mandatory paid AI	Forbidden by current default; paid budget = 0.
Microservices	No extraction without measured need + ADR.
Full A1-C1 simultaneous content drop	Curriculum scales module-by-module after reference baseline.
Public launch/legal geography	Separate pre-launch decisions on brand, hosting region, minors/privacy and production policies.

2. Незмінні implementation invariants

Invariant	Implementation consequence	Blocking proof
One shared engine	English/Czech differences are config/evaluator/locale strategy, never separate app forks.	Czech M01 runs without English-only workaround.
PostgreSQL source of truth	Progress, attempts, evidence, mastery, review, publications and AI policy/audit are durable.	Cache/provider outage does not alter learning state.
Immutable evidence/publications	Attempts/evidence and published snapshots are versioned; rollback changes pointers.	Historical replay/rollback test.
Two-level mastery	6 global skills + knowledge-item facets; Review schedules at knowledge/facet level.	Facet and skill projection tests.
Server-authoritative evaluation	Client/AI cannot set correct/mastery/due_at.	Tamper/provider-disagreement tests.
Pronunciation safe binary UX	Usable speech -> correct/try again; unavailable/low confidence neutral and non-blocking.	Five-path speech fixture.
Pre-generated media	Published lesson binds asset_id + asset_version.	Provider-offline manifest preflight.
AI optional/provider-neutral	Core course works with all AI disabled; provider SDKs stay in adapters.	Provider-offline EN+CS E2E.
Paid AI budget 0	No silent billable fallback.	Pre-call budget guard test.
Original content/design	No proprietary Babbel content/media/trade dress copied.	Originality/provenance review.

3. Delivery strategy і critical path

Primary strategy
Build the smallest production-shaped system that proves the hardest shared contracts with real English and Czech content. Freeze that reference baseline only after cross-language, mastery, speech, media, accessibility, privacy and AI-off regression evidence passes.

```
P0 Governance freeze
  -> P1 Repository/CI foundation
    -> P2 Data + language isolation + publication
      -> P3 Lesson Engine + attempts + deterministic state
        -> P4 Two-level Mastery + Review + remediation
          -> P5 Media + audio + async speech
            -> P6 English M01 vertical slice
              -> P7 Czech M01 without engine fork
                -> P8 Admin/CMS + publication + shared AI foundation + Admin AI Assist
                  -> P9 Optional learner AI + AI Auto allowlist (reuses P8 AI foundation)
                    -> P10 Product surfaces + settings
```

```
                    -> P11 Reference hardening + baseline freeze
                      -> P12 A1 M02-M12 rollout + pilot/calibration + optional Level Test
                        -> P13 Full-system hardening + launch/no-launch gate
```

3.1. Parallel tracks

Track	May start	Must wait for
Content briefs / knowledge inventory	After P2 schema/ID model is stable enough for fixtures.	Bulk publication waits for P11.
Media requirements/prompts/scripts	After activity/media contracts are stable in P3/P5.	Final binding waits for QA-approved asset versions.
English/Czech linguistic review	As soon as draft fixtures exist.	Publish waits for formal review gates.
Admin UI	Can begin after P2/P3 API contracts.	Publish controls require P8 workflow services.
AI provider spike	Can run in parallel after P1 with synthetic data.	Learner AI rollout waits for P9 policy/grounding/budget gates.
M02-M12 authoring	Can prepare briefs after P3/P4 contracts stabilize.	Full production scaling waits for P11 reference freeze.

4. Repository, branching i environments

4.1. Repository baseline

```
lingvessa/
  apps/
    web/       learner PWA
    admin/     Admin/CMS
    api/       NestJS modular monolith
    worker/    media/speech/AI/publication/privacy jobs
  packages/
    domain/
    application/
    activity-schemas/
    lesson-engine/
    mastery-review/
    ai-core/
    provider-adapters/
      ai/
      speech/
      tts/
      image/
      email/
      storage/
    api-client/
    ui/
    content-tools/
  content/
    english/
    czech/
    shared/locales/uk-UA/
  assets/specs/
  prisma/
  infra/
  docs/adr/
  ai/prompts/
  ai/policies/
  ai/evals/
```

- Private Git repository is source of truth for code, schemas, migrations, tests, prompts/policies, manifests and docs.
- Generated media bytes and production secrets are not normal Git artifacts; store media in object storage, secrets in runtime secret management.
- Main branch protected; implementation moves through reviewed PRs/commits. Every milestone has a tagged/recorded baseline.
- Behavior-changing contract changes require ADR/spec revision before merge; do not bury architecture changes in feature code.

4.2. Environment order

Environment	Use in plan	Gate
local	P1 onward; synthetic EN/CS fixtures and local services.	One-command bootstrap + deterministic tests.
preview	Per PR when deployment wiring is ready.	Protected; synthetic data; auto-expiry where supported.
development/reference	Current low-cost deployment: GitHub Private + Northflank Sandbox + R2.	No production SLA assumptions; portability preserved.
staging	Required before P11 freeze and release drills.	Migration, rollback, provider failure, backup restore, AI evals.
production	Not required for reference implementation.	Explicit hosting/region/privacy/legal/ops ADR before public use.

5. Workstreams та ownership boundaries

Workstream	Owns	Key outputs
Platform foundation	repo, build, config, CI/CD, containers, migrations	reproducible build + environment contract
Identity/language	users, sessions, language profiles, ownership, isolation	EN/CS switching and IDOR-safe state
Content/publication	schemas, knowledge, versions, publications, approvals	immutable manifests + rollback
Learning runtime	LessonPlayer, activities, attempts, state, resume	deterministic end-to-end lesson flow
Mastery/Review	evidence, facets, skills, scheduler, mistakes, collections	recomputable projections + due queues
Media/speech	asset pipeline, audio, voice, mic, speech adapter	normal/slow audio + safe pronunciation UX
Admin/CMS	authoring, validation, review, publish/rollback, audit	manageable content lifecycle
AI	policy, routing, prompts, grounding, evals, usage/budget	optional learner/admin capabilities
QA/security/ops	tests, a11y, privacy, observability, restore	release evidence + defect control
Curriculum/content	lesson briefs, knowledge inventories, dialogues, media specs	approved EN/CS learning content

6. Milestone map P0-P13

Milestone	Primary result	Hard exit evidence
P0 Governance freeze	Exact baseline + decision log + full MUST trace matrix	P0 closed; P1 allowed
P1 Repository foundation	Private monorepo, CI, typed config, package map	Clean checkout/build/test
P2 Data/isolation/publication	UUID/stable_code/version_code + composite isolation + immutable publication	DB negative isolation + manifest tests
P3 Learning core	Lesson Engine, attempts, deterministic state, resume/remediation	Reference fixture E2E
P4 Mastery/Review	Canonical facets, six skill projections, review/mistakes/collections	Recompute + no cross-language review
P5 Media/Speech	Assets/audio/voices + async speech_job lifecycle	Media/speech failure tests
P6 English M01	E-A1-M01 + Recap	English vertical-slice QA
P7 Czech M01	C-A1-M01 + Recap on same engine	Czech + isolation QA
P8 Admin/CMS + shared AI foundation + Admin AI Assist	Draft/review/publish/rollback + AI Assist foundation	Publication drill + AI Assist cannot bypass gates
P9 Learner AI + AI Auto allowlist (reuses P8 AI foundation)	Opt-in learner AI, provider routing, AI Auto low-risk only	AI-off + opt-in + quota/privacy tests
P10 Product surfaces/settings	Onboarding/Home/Explore/Course/ Progress/Profile/Settings/notifications	Full settings persistence + DST/quiet-hours E2E
P11 Reference hardening	32 scenarios + technical/a11y/privacy/security/restore gates	Signed reference release-candidate report
P12 A1 rollout + pilot/Level Test	M02-M12 controlled rollout, pilot/calibration, optional A1 Level Test	Coverage/pilot/calibration report
P13 Full-system hardening/launch	Security/privacy/performance/restore/DR/ release provenance/legal-product readiness	All launch gates green or explicit no-launch decision

7. P0 - Baseline lock and implementation readiness

- Record the exact seven-document baseline (six governing specifications + this Implementation Plan) in repository docs/baseline.
- Create decision log for fixed invariants and list only truly open decisions.
- Create requirement trace matrix: Formal / Technical Specification / Technical Design / Curriculum / Reference / AI MUST -> backlog / test / ADR / owner. Curriculum CUR-A1-* is explicitly included.
- Confirm reference modules are the first executable acceptance fixtures.
- Define issue labels, PR template, change-control rule and Definition of Ready.

Контроль	Вимога
Exit evidence	baseline.yml or equivalent + decision log + trace matrix

Контроль	Вимога
Blocker	any unresolved contradiction affecting data/state/mastery/speech/AI authority
Not required	public brand clearance, production hosting or paid AI

8. P1 - Repository and engineering foundation

- Create private monorepo and exact runtime/tooling lockfile.
- Create apps/web, apps/admin, apps/api, apps/worker and required packages with dependency lint boundaries.
- Add typed environment config, secret scanning, lint/typecheck/unit-test/build pipelines.
- Create local PostgreSQL bootstrap, initial Prisma migration and synthetic bilingual fixtures.
- Create OCI container build and health/readiness endpoints.
- Prepare development deployment config without embedding Northflank/R2 assumptions into domain code.

Контроль	Вимога
Exit evidence	fresh clone -> install -> DB migrate -> test -> build succeeds
Blocker	provider SDK or framework leaks into domain package; secrets in repo
Deployment	local mandatory; remote dev optional but preferred after clean local build

9. P2 - Data, language isolation and publication foundation

- Implement users/sessions/roles/consents minimal foundation and language_profiles.
- Implement curriculum/content/knowledge IDs and typed knowledge edges.
- Implement content_versions, publications, approvals and immutable publication manifest.
- Implement language/variety scope: uk-UA UI; en-GB default/en-US selectable; cs-CZ separate.
- Implement database guards against cross-language attempts/mastery/review/collections/AI sessions.
- Implement content schema validator and reference fixtures loader.

Контроль	Вимога
Exit evidence	schema/migration tests + EN->CS->EN isolation + immutable publication fixture
Blocker	cross-language reference possible; content identity tied to filenames/titles/provider URLs
Freeze	stable IDs and schema major before P3 content authoring expands

10. P3 - Lesson Engine and deterministic learning core

- Implement LessonPlayer lifecycle and activity registry by type + schema version.
- Implement core M01 renderers/evaluators: intro, choice, matching/classification, gap, builder, typing, interactive grammar, listening, dialogue chain, role-play/text fallback, result/remediation.
- Implement scored state machine: correct-or-reveal, retry, assisted evidence, Auto-advance only from allowed completion states.
- Implement server-authoritative attempt command with idempotency and transactional outbox.
- Implement lesson run persistence, reload/login resume, client unsent queue and version conflict behavior.
- Implement remediation loop for scored mistakes and stable error codes.

Контроль	Вимога
Exit evidence	synthetic EN and CS lesson fixture E2E; duplicate attempt does not duplicate evidence
Blocker	client can force correct/mastery; reveal counted as independent success; dead-end state
AI rule	no AI call is required anywhere in P3

11. P4 - Two-level Mastery, Review, mistakes and collections

- Implement immutable evidence_events mapped to skill and relevant facet.
- Implement knowledge_mastery_states and derived six skill_mastery_states.
- Implement deterministic/versioned mastery update and recompute path.
- Implement review_schedules keyed by knowledge + facet + scheduler version.
- Implement due/weak/error session builder, spacing/deduplication and reveal/hint scheduling effects.
- Implement mistake instances, lesson remediation link, reset semantics and personal collections isolated by language.

Контроль	Вимога
Exit evidence	recompute/golden tests; meaning/spelling/listening/register remain distinct
Blocker	single percentage as source of truth; cross-language due item; delete collection deletes mastery
UX	full skill profile lives in Progress; lesson result shows relevant indicators only

12. P5 - Media, audio and pronunciation

- Implement media requirements, jobs, assets, asset_versions, voice/character bindings and manifest preflight.
- Integrate S3-compatible object storage; development default may use Cloudflare R2.
- Generate/prepare original reference images, avatars, vector assets and audio scripts; record provenance/checksums/alt/transcripts.
- Implement normal + slow playback and multi-speaker dialogue synchronization.
- Implement microphone permission and pulsing recording UX with reduced-motion alternative.
- Implement speech adapter states: speech_correct / speech_incorrect / speech_unavailable; raw score hidden from learner.
- Keep raw microphone bytes transient by default.

Контроль	Вимога
Exit evidence	all M01 required asset IDs resolve; speech correct/mismatch/denial/low-confidence/provider-failure fixtures pass
Blocker	lesson requires live generation; speech failure blocks progress; raw audio stored unintentionally
Variety	voice metadata and accepted variants correctly distinguish en-GB/en-US/cs-CZ

13. P6 - English reference module E-A1-M01

- Load all four English lessons and integrated Recap from Reference Module Design.
- Bind approved media and voices; verify en-GB default and en-US meaningful variants.
- Verify be/a-an/contractions/origin/spelling/repair phrase evaluation and mastery facet mapping.
- Verify lesson results, remediation, review seed, reload/resume and no hard speech gate.
- Run English subset of reference regression suite on mobile and desktop viewports.

Контроль	Вимога
Exit evidence	English vertical-slice evidence report + defect list
Blocker	hardcoded lesson content in renderer; incorrect accepted variant; unresolved media
Decision	contract defect changes shared engine/spec, not English-only workaround

14. P7 - Czech reference module C-A1-M01

- Load all four Czech lessons and Recap without creating a Czech-specific engine fork.
- Implement/verify ty/vy register, být forms, pronoun omission, origin forms, Czech diacritics, Unicode NFC and strict vs meaning-focused policies.
- Verify first-syllable stress, vowel length, ř awareness and Czech voice/audio metadata.
- Verify form errors do not erase meaning mastery and word order is not reduced to English token rules.
- Run cross-language switch and review-isolation tests with both reference modules populated.

Контроль	Вимога
Exit evidence	Czech vertical-slice evidence report + no-engine-fork architecture test
Blocker	English-only schema workaround; global diacritic stripping; register collapse
Reference gate	P6+P7 jointly prove bilingual engine viability

15. P8 - Admin/CMS and publication workflow

- Implement structured authoring/editing around content versions and validation reports.
- Implement linguistic/pedagogical/media/accessibility/technical approval gates.
- Implement media requirement/job inspection and approved asset binding.
- Implement snapshot publish, rollback pointer switch, audit trail and checksum-sensitive approvals.
- Implement coverage/gap reports and preview for core activity types.
- Implement the shared AI foundation required by Admin Assist before any real AI call: AI Gateway/Orchestrator, Policy Engine, Provider Router, provider-neutral adapter contract plus at least one approved free/local adapter, Prompt Registry, Output Validator, Usage Meter and audit. Admin AI Assist may call providers only through this foundation; AI Auto remains OFF and privileged actions remain manual.

Контроль	Вимога
Exit evidence	draft -> validate -> review -> media -> publish -> rollback staging drill
Blocker	editor/worker can directly mutate published snapshot; rollback rewrites attempts
Admin rule	privileged actions audited; service identities least-privileged

16. P9 - Optional AI Layer

- Reuse the shared AI foundation delivered in P8; add learner-specific Grounding Builder/context minimization, learner capability policies and conversation/session services. Do not duplicate gateway/router/policy components.
- Extend the P8 provider adapters/configuration only as needed for learner capabilities; learner AI reuses the same Gateway/Policy/Router/Validator/Usage/Audit foundation rather than creating a second stack.
- Enforce free_only=true, paid_inference_enabled=false and paid budget=0 before any provider call.

- First learner capabilities: grounded "Поясни інакше" and bounded hints; then optional extra practice, writing feedback and text role-play by flags.
- Learner AI is implemented only after Admin AI Assist/policy/audit foundation. AI Auto remains opt-in allowlist for draft/low-risk actions only.
- Prohibit direct AI publish/rollback/account delete/role/billing; all tool actions re-enter normal application authorization.
- Add provider-offline, 429/5xx, invalid-output, grounding and cost guard tests.

Контроль	Вимога
Exit evidence	AI contract/eval suite + EN/CS grounding fixtures + core E2E with every AI provider disabled
Blocker	AI output directly changes mastery/publish; hidden paid fallback; broad data export to provider
Rollout	AI failure always has authored/scripted/non-AI fallback

17. P10 - Product surfaces and settings

- Onboarding: create/select a real English or Czech language profile; switching preserves independent state and never creates a placeholder course.
- Home: show current course, resume entry, due Review, progress/streak summary and relevant recommendations with no AI dependency.
- Explore/Course: implement level/module/lesson navigation, prerequisite-gap warnings and manual level access without a hard CEFR lock.
- Progress: show exactly six global skill projections plus course/level progress; any Level Test Reading result is assessment-only and is not a seventh mastery skill.
- Profile/Settings: persist target language, en-GB/en-US, UI locale, Auto-advance, audio/speech preferences, learner AI opt-in/consent, lesson/module/all reset and account deletion controls.
- Notifications/quiet hours: persist enablement, IANA timezone and quiet-hours settings; verify DST-safe scheduling and that notifications can be disabled without affecting learning.

Exit evidence	Onboarding/Home/Explore/Course/Progress/Profile/Settings/notifications E2E; settings survive reload/login; language switching preserves independent state; quiet-hours scheduling is DST-safe.
Blocker	Settings lost/not persisted, learner AI opt-in ignored, cross-language profile mutation, or notifications violate disable/quiet-hours policy.
P10 gate	P10 closes product surfaces/settings persistence only; it does not freeze the reference baseline.
Exit to P11	All P10 product-surface and settings-persistence gates pass.

18. P11 - Reference hardening and baseline freeze

- Run the complete 32-scenario Reference Module regression suite plus Technical Design acceptance scenarios for English, Czech and cross-language isolation.
- Run responsive/mobile/desktop visual QA; keyboard/touch/accessibility; transcript/non-mic paths; reduced motion; settings persistence and product-surface E2E.
- Run security/privacy tests: IDOR, roles, microphone lifecycle, learner-AI opt-in, AI payload allowlist, secrets/logs, retention and account deletion/export paths.
- Run publish/rollback, migration compatibility, provider failure, queue/outbox and backup/restore drills in staging; verify release provenance and immutable history.
- Enable observability for errors, latency, content versions, queue/provider failures, AI quota/cost/fallback and correlation IDs; define release-blocking alert ownership.
- Resolve shared-contract defects through ADR/spec revision, rerun regression and freeze the bilingual reference baseline only after all P11 gates pass.

Exit evidence	Signed P11 reference hardening report: all 32 reference scenarios + technical/a11y/privacy/security/restore/release-provenance gates pass, with zero unresolved release blockers.
Freeze result	E-A1-M01 and C-A1-M01 become frozen executable regression fixtures for subsequent content and engine changes.
Blocker	Any unresolved shared-contract, security/privacy, restore, accessibility or language-isolation defect blocks P11 closure and P12.
Exit to P12	Reference baseline is frozen only after the full P11 regression/hardening report passes.

19. P12 - A1 rollout M02-M12, pilot/calibration and optional A1 Level Test

- Approve/freeze both 12-module A1 maps and module outcomes, then author M02-M12 lesson briefs, Recap briefs, recycle plans and knowledge inventories for English and Czech.
- Produce lesson content and original media in controlled bilingual batches through the approved schema, language, pedagogy, media, accessibility and publication gates.
- Collect pilot evidence for completion, errors, reveals, Review outcomes, latency, drop-off, scheduler behavior, speech false negatives and optional-AI usefulness; change rules only through versioned decisions.
- Build and calibrate optional A1 Level Test forms only after sufficient content/pilot evidence; Reading may be a separate assessment subscore but never a seventh mastery skill, and no pass threshold is invented in advance.
- Keep A1->A2 access non-blocking and close P12 only with coverage/prerequisite reports, pilot/calibration decisions and a controlled rollout record ready for P13.

Exit evidence	QA-approved M02-M12 publications with coverage/asset manifests plus pilot/calibration report; optional A1 Level Test forms/rubrics are versioned only if calibrated.
Batch rule	Small recoverable bilingual batches; immutable publication versions; no giant unreviewed generation job.
Calibration	Lexical/readiness targets remain planning targets until evidence supports a versioned decision; Reading is assessment-only, not a seventh mastery skill.
Next	after stable A1, expand A2-C1 and professional tracks incrementally

20. P13 - Full-system hardening and launch gate

- Full security and authorization suite, secret/supply-chain scanning and admin MFA checks.
- Performance/capacity budgets under representative A1 catalog and review load.
- Backup restore, point-in-time recovery where available, object-store restore/version verification and disaster-recovery drill.
- Privacy/data deletion/export/retention checks including AI sessions and temporary speech audio.
- Release provenance links commit, container digest, migrations, content/media manifests, AI policies/prompts and approvals.
- Cross-browser/mobile accessibility regression and full EN/CS language-isolation E2E.
- Public launch remains blocked until product/legal/brand/hosting-region/minor policy decisions are explicitly approved.

21. Cross-cutting QA and release gates

Gate	Runs from	Release-blocking checks
Schema/contract	P1	JSON schema, OpenAPI, migrations, fixtures, backward/forward compatibility.
Language correctness	P3	en-GB/en-US/cs-CZ policies, Czech diacritics/register/morphology, accidental Cyrillic lint.
Learning state	P3	correct-or-reveal, idempotency, immutable evidence, resume, no dead ends.
Mastery/Review	P4	skill/facet separation, recompute, scheduling effects, language isolation.
Media/speech	P5	asset resolution, provenance/rights, normal/slow, transcript/alt, safe speech fallback.
Reference E2E	P6	English M01 all core flows.
Cross-language E2E	P7	Czech M01 same engine + switching/isolation.
Publication/admin	P8	approval, checksum, publish/rollback, audit, least privilege.
AI foundation/Admin Assist	P8	Gateway/Policy/Router/Validator/Usage Meter, provider swap, zero-paid-budget, admin proposal/diff/audit, no authority bypass.
Accessibility	P3-P13	keyboard/touch/focus, text alternatives, reduced motion, non-mic path.
Security/privacy	P1-P13	IDOR/RBAC, secrets, audio/AI data minimization, retention/deletion.
Reliability/ops	P8-P13	backup/restore, migration rollback, provider fault, observability.
Originality	content milestones	provenance and review against copied proprietary content/trade dress.
Learner AI	P9	Opt-in/consent, grounding, explain/hint/practice/writing/conversation fallback, privacy/quota/retention, AI Auto allowlist.

21.1. Required release evidence package

- Git commit SHA + container/image digest + DB migration set.
- Content publication manifest + asset manifest/checksums.
- Schema/OpenAPI/fixture versions + evaluator/mastery/scheduler versions.
- Test report: unit/integration/E2E/regression/a11y/security relevant to release.
- Approval records for content/media and AI policies where used.
- Known limitations and rollback target.
- AI provider/prompt/policy versions and budget/fallback configuration for AI-enabled release.

22. Content/media parallel production plan

Rule
Do not wait until the platform is finished to start all content, but do not bulk-produce content before the contracts it depends on are stable. Content work is staged against schema maturity.

Stage	Content work allowed	Not yet allowed
P0-P1	Refine existing M01 fixtures and inventory.	Bulk M02-M12 production.
P2	Assign stable knowledge/outcome IDs; draft module briefs.	Final evaluator-specific activities before schema freeze.
P3-P4	Draft M02-M12 lesson briefs, recycle/facet maps; author text-only fixtures.	Mass media generation.
P5	Create media requirements, prompts/scripts and controlled media batches.	Publish unresolved/unreviewed assets.
P6-P7	Use real reference lessons to validate authoring patterns and quality.	Copy reference lesson mechanically into later modules.
P8	Move repeatable authoring/review/publish work into Admin/CMS; shared AI foundation is live; Admin AI Assist may create drafts/QA/media specs through policy.	Bypass human/policy gates; learner AI exposure before P9 opt-in/grounding gates.
P9	Enable optional learner AI explain/hint/practice/writing/conversation and low-risk AI Auto using the already-tested P8 foundation.	Treat AI output as curricular truth, direct publish, implicit paid fallback, or learner AI without opt-in.
P12	Scale M02-M12 in controlled bilingual batches.	One giant unreviewed course generation job.

23. Backlog and task packet contract

23.1. Every implementation task must contain

Field	Required content
Task ID	e.g. IMP-P3-ATT-001
Governing refs	Exact spec requirement IDs / design section / reference fixture.
Goal	One verifiable behavior, not broad "build backend" wording.
Dependencies	Merged tasks/schema/migration/content prerequisite.
Files/modules	Expected ownership boundary; no unauthorized cross-module writes.
Acceptance	Automated test or explicit QA evidence.
Negative test	How failure/abuse/provider outage is tested where applicable.
Migration/compatibility	Required if schema/API/content contract changes.
Observability	Event/log/metric/correlation needed to debug the behavior.
Rollback	How change is disabled/reverted without rewriting evidence/history.

22.2. Definition of Ready

- No unresolved product ambiguity that changes the governing contract.
- Dependent schema/interface is versioned or explicitly part of the same task.
- Acceptance test is written before or together with implementation.
- Reference fixture/data exists or is included in task.
- Any external provider dependency has adapter boundary and fallback behavior defined.

22.3. Definition of Done for a task

- Code + tests + migrations/schema updates are committed together.
- Clean checkout passes required checks.
- No secret/provider-specific domain leakage.
- Failure and retry behavior verified; no dead end.
- If behavior changes a shared contract, ADR/spec/version is updated before milestone closes.
- Relevant reference regression tests remain green.

24. Risk, decision and change control

Risk	Implementation response	Decision deadline
Overengineering	Keep modular monolith; only measured extraction.	Before any microservice proposal.
English-first bias breaks Czech	P7 is hard architectural gate; no Czech fork.	Before P11 freeze.
Speech false negatives	Confidence gate + neutral unavailable + calibration.	Before speech production rollout.
AI hallucination	Grounding + schema/rubric + optionality + evals.	Before each learner capability flag.
Free provider disappears	Adapters + provider registry + AI-off fallback.	Continuous; no permanent free-tier assumption.

Risk	Implementation response	Decision deadline
Accidental AI billing	Server budget gate; default 0.	Before any provider call.
Media inconsistency	Manifest + provenance + QA + versioned asset IDs.	Before each publication.
Content scale before engine stability	Reference freeze before bulk M02-M12.	P11 gate.
Data/privacy risk	Minimum outbound AI context; transient mic audio; audited lifecycle.	Before public learner AI/voice use.
Brand/public launch uncertainty	Keep Lingvessa as working name until clearance.	Before public branding/launch.

23.1. Change rule

Controlled revision
Any change to language isolation, state/completion contract, mastery/evidence mapping, AI authority/permissions, paid budget policy, publication immutability, media binding or privacy defaults requires a reviewed spec/ADR/migration/test update. Historical evidence and published snapshots are never overwritten to make the system appear current.

25. Implementation Definition of Done

- Private monorepo builds reproducibly; environments use typed config and no secrets in source/client/logs.
- English and Czech reference modules run through one shared versioned content schema and engine.
- Deterministic attempts, correct-or-reveal, remediation, resume and idempotency work with immutable evidence.
- Two-level mastery persists knowledge-item facets and derives six global skills; Review is facet-aware and language-isolated.
- All required reference media is system-produced/pre-generated, versioned, approved and accessible.
- Pronunciation UI implements pulsing mic plus usable correct/try-again and neutral speech_unavailable without hard gate.
- Admin/CMS supports structured authoring, validation, approvals, publish/rollback, audit and media jobs.
- Optional AI Layer is provider-neutral, grounded, free/local-first, paid budget 0 by default and cannot bypass mastery/publish/privileged controls.
- Core English/Czech learning, Review and progress complete with all AI providers disabled.
- 32 reference regression scenarios plus technical/a11y/security/privacy/restore gates pass.
- Reference baseline is frozen and reusable as a regression fixture before M02-M12 scaling.
- No copied proprietary course content/assets/trade dress is present in production artifacts.

Appendix A. Dependency matrix

Milestone	Depends on	Unlocks
P0	current documents	P1
P1	P0	P2 + provider spikes
P2	P1	P3 + controlled content authoring
P3	P2	P4 + broader lesson fixtures
P4	P3	P5 + facet-aware content maps
P5	P3/P4	P6/P7 complete media/speech
P6	P3-P5	English proof
P7	P3-P6	shared bilingual proof
P8	P2-P7	scalable content lifecycle
P9	P3/P4/P8	optional AI capabilities
P10	P6-P9	P11
P11	P1-P10	P12
P12	P11	P13
P13	P12	launch/no-launch decision

Appendix B. First coding backlog

Order	Task packet	Minimum acceptance
1	IMP-P0-001 Record baseline + trace matrix	All six governing specifications, including Curriculum CUR-A1-* MUST requirements, are recorded and mapped to backlog/test/ADR/owner; decision log approved.
2	IMP-P1-001 Initialize private pnpm monorepo	Clean install/typecheck/test/build.
3	IMP-P1-002 Add typed config + secret scanning	Invalid config fails fast; no secrets in bundle.
4	IMP-P1-003 Add PostgreSQL/Prisma bootstrap	Migration and seed repeatable.
5	IMP-P1-004 Create apps/web, admin, api, worker + package boundaries	Dependency lint passes.
6	IMP-P2-001 Implement language_profiles + isolation keys	EN/CS switch preserves independent state.
7	IMP-P2-002 Implement content/knowledge/publication schema	Immutable publication fixture and validator.
8	IMP-P3-001 Implement lesson manifest loader + registry	EN/CS fixture renders same activity type.
9	IMP-P3-002 Implement attempt transaction/idempotency	Duplicate request credited once.
10	IMP-P3-003 Implement scored state machine + remediation	Wrong->retry/reveal->Next; reveal assisted.
11	IMP-P4-001 Implement evidence/facet projections	Meaning vs spelling vs pronunciation separate.
12	IMP-P4-002 Implement review scheduler/session builder	Due/weak/error; no cross-language item.
13	IMP-P5-001 Implement media manifest/object adapter	Concrete asset version resolves.
14	IMP-P5-002 Implement audio/slow/voice binding	Normal/slow + multi-speaker works.
15	IMP-P5-003 Implement microphone/speech state machine	correct/incorrect/unavailable paths all safe.
16	IMP-P6-001 Load E-A1-M01	English vertical slice E2E.
17	IMP-P7-001 Load C-A1-M01	Czech vertical slice without engine fork.
18	IMP-P8-001 Minimal Admin content/version/publish flow	Draft->approved->publish->rollback.
19	IMP-P8-002 Shared AI foundation: Gateway/Policy/Router/Validator/Usage Meter + provider doubles	Foundation works with paid budget 0; provider policy, structured validation, audit and no-authority bypass tests pass before Admin Assist calls.

Order	Task packet	Minimum acceptance
20	IMP-P8-003 Admin AI Assist draft/QA path	Admin Assist uses P8 foundation, produces draft/proposal only, and cannot bypass publish/role/budget gates.
21	IMP-P9-001 Learner AI explain/hint + opt-in/fallback	Learner AI remains OFF by default; opt-in, grounding, quota/privacy and AI-off authored fallback tests pass.
22	IMP-P9-002 Learner practice/writing/conversation + AI Auto allowlist	Practice evidence gates, conversation retention, writing schema fallback and low-risk AI Auto allowlist pass; paid budget remains 0.
23	IMP-P10-001 Product surfaces/settings persistence E2E	Onboarding/Home/Explore/Progress/ Profile/Settings and notification/quiet-hours/DST persistence gates pass, including learner AI setting persistence.
24	IMP-P11-001 Full reference regression run	Run only after P1-P10 gates pass; all 32 scenarios green or defects explicitly resolved/versioned before P11 freeze.

Appendix C. Handoff checklist

- Current baseline PDF set is attached/available to the coding context.
- Implementation Plan is treated as execution order, not permission to override governing specs.
- Before each milestone, read only the governing sections needed for that milestone plus reference fixtures.
- One milestone is implemented in bounded PR/task packets; do not attempt the entire platform in one code generation pass.
- After each milestone, run automated tests and inspect actual UI/flows before starting the next dependency.
- If a defect exposes a contract gap, stop and version the contract/ADR instead of patching a language-specific workaround.
- Keep generated media and provider credentials outside Git; commit manifests/specs/checksums instead.
- Admin AI Assist may run only after the P8 shared-AI-foundation and Admin Assist gates pass. Keep learner AI and P9 AI Auto capabilities disabled until their P9 controls are proven. Paid inference remains disabled and the paid budget remains 0 unless separately approved.
- Do not start bulk A1 generation until P11 reference baseline is frozen.
- Public launch is a separate decision after technical/product/legal readiness, not an automatic consequence of P12.

Approval record

Поле	Значення
Document	Lingvessa Implementation Plan v1.4
Created	09.09.2026
Approval	Governance approval record: approver, approved_document_version, approval_date, decision, accepted_exceptions. Runtime CMS Approval is a separate content entity defined in Technical Specification/Design.
Execution start condition	Start with P0 governance materialization in the repository. P1 may begin only after /docs/baseline decision log and requirement-trace links are committed and reviewed.
Change policy	Future execution-order or gate changes create v1.4+; do not overwrite this v1.4.

Final execution rule
The first real coding target is not the whole A1 course and not AI. It is a production-shaped bilingual reference foundation that can load, evaluate, persist, review and render English and Czech M01 reliably. Everything else scales from that verified base.

Controlled baseline synchronization - 09.09.2026

Normative baseline
This document is part of the synchronized Lingvessa owner-locked implementation baseline; P0 is authorized as of 10.09.2026. Earlier current-version references are superseded by the exact set below; historical "replaces" statements remain historical only.

Document	Version	Authority
Lingvessa_Formal_Specification	v1.7	Product scope, pedagogy, user-facing invariants
Lingvessa_Technical_Specification	v1.7	Normative technical requirements and interfaces
Lingvessa_Technical_Design	v1.7	Concrete implementation contracts, DDL/state/API design
Lingvessa_Curriculum_Blueprint_A1	v1.6	A1 pedagogical scope and module architecture
Lingvessa_Reference_Module_Design	v1.6	Executable E-A1-M01/C-A1-M01 fixtures
Lingvessa_AI_Integration_Specification	v1.4	AI-specific policy, data, routing, permissions
Lingvessa_Implementation_Plan	v1.4	Sole execution-order authority; no scope override

Decision	Topic	Fixed rule
BL-001	Baseline set	Formal 1.7 / Technical 1.7 / Design 1.7 / Curriculum A1 1.6 / Reference 1.6 / AI 1.4 /

		Plan 1.4
BL-002	Precedence	Domain-specific governing document controls; unresolved conflict -> owner decision + ADR/spec revision before code merge.
BL-003	Identity contract	id = UUID; stable_code = immutable human-readable semantic code; version_code = immutable revision label. Never put stable_code in *_id.
BL-004	Mastery contract	Exactly six global skills; canonical facet registry in this release. Reading is assessment-only domain, not seventh mastery skill.
BL-005	Language isolation	Composite DB FKs/checks/trigger guards enforce user + language_profile + language lineage for learner state and learner AI.
BL-006	Pronunciation states	speech_correct / speech_incorrect / speech_unavailable are canonical. Incorrect is non-blocking; Auto-advance occurs only after feedback dwell and never records success.
BL-007	Speech lifecycle	Asynchronous speech_job lifecycle with idempotent final attempt settlement and transient audio cleanup.
BL-008	AI API/data model	Canonical /api/v1/ai/explain|hint|practice|writing-feedback|conversations plus unified ai_* physical model.
BL-009	Monorepo	One /packages/provider-adapters package with ai/, speech/, tts/, image/, email/, storage/ sub-adapters; /api-client and /ui retained.
BL-010	Execution order	Implementation Plan v1.4 P0-P13 is the sole milestone numbering authority.
BL-011	Learner AI opt-in	Learner AI is OFF by default until explicit opt-in/consent. Effective enablement order is global -> capability -> provider/policy -> learner setting/consent -> quota.
BL-012	OpenAPI	Pin API description to OpenAPI Specification 3.1.2 for v1 baseline; upgrades require ADR/contract diff.

Precedence: Formal controls product/pedagogical scope; Curriculum controls A1 pedagogy; Technical Specification controls normative technical requirements; Technical Design controls concrete implementation detail; AI Specification controls AI-specific behavior; Reference Module Design is the executable fixture; Implementation Plan controls execution order only. Any unresolved conflict blocks implementation until an owner decision and controlled revision/ADR.

Canonical mastery/facet registry

Machine-code rule
Only the codes in this registry may appear in mastery_facet/evidence_map. Human labels such as “gist”, “detail”, “target feature”, “repair”, or directional recall must map to these exact machine codes.

Facet code	Meaning
recognition	Recognition of a known form/choice
meaning	Meaning comprehension
recall_l2_to_l1	Target-language -> explanation-language recall
recall_l1_to_l2	Explanation-language -> target-language recall
production	Independent target-language production
spelling	Spelling of lexical/form surface
orthography	Language-specific orthographic correctness
listening_recognition	Recognition from audio
listening_gist	Gist from audio
listening_detail	Specific detail from audio
pronunciation_target	Named pronunciation feature
intelligibility	Understandability of learner speech
context_use	Appropriate use in context
collocation	Chunk/collocation integrity
grammar_form	Grammar form
word_order	Word order
morphology	Inflection/morphology
register	Register/formality
variant_awareness	en-GB/en-US or accepted language variant awareness
interaction_repair	Repair/clarification move
interaction_goal	Goal completion in interaction

Facet-to-global-skill aggregation

Global skill	Eligible facets (only when evidence_map explicitly targets the skill)
Vocabulary	recognition, meaning, recall_l2_to_l1, recall_l1_to_l2, production,

	spelling, context_use, collocation, variant_awareness, register
Grammar	grammar_form, morphology, word_order, register, context_use
Listening	listening_recognition, listening_gist, listening_detail, meaning
Speaking	production, context_use, register, interaction_repair, interaction_goal, intelligibility
Writing	production, spelling, orthography, grammar_form, context_use, register
Pronunciation	pronunciation_target, intelligibility

Reading is not a seventh global mastery skill. Reading tasks may emit item-facet evidence to Vocabulary and/or Grammar according to the objective. The optional Level Test may report a separate assessment_domain=reading subscore; that subscore is assessment-only and is not persisted as skill_mastery_states.

Canonical identity and version contract

Field	Type	Example	Rule
id	UUID	018f...	Opaque database/API identity.
stable_code	string	E-A1-M01-L01	Immutable human-readable semantic code; never stored in *_id.
version_code	string	1.4 / 2026-09-05.1	Immutable revision label for a versioned entity.
provider_external_id	string nullable	vendor-model-x	External/provider identifier only in adapter/config boundary.

API/JSON examples MUST use UUID-shaped values in *_id fields. Content manifests MAY use stable_code for authoring references, but the build/publish step resolves them to UUID + version_code and rejects unresolved codes.

Database-enforced English/Czech isolation

Isolation invariant Service-layer checks are not sufficient. Learner-state tables denormalize language_id where needed so PostgreSQL composite foreign keys can prove lineage.
language_profiles: UNIQUE(id,user_id,language_id), UNIQUE(user_id,language_id). lessons/publications/content_versions/activity_items/knowledge_items: UNIQUE(id,language_id). lesson_runs: explicit user_id,language_profile_id,language_id; composite FK to language_profiles/publication/lesson. activity_attempts: explicit user_id,language_profile_id,language_id; UNIQUE(id,user_id,language_profile_id,language_id); composite FK to lesson_runs; item/content FKs carry language_id. evidence_events: explicit user_id,language_profile_id,language_id; composite FK(attempt_db_id,user_id,language_profile_id,language_id)->activity_attempts; knowledge/content FKs carry language_id; facet requires knowledge_item_id. ai_runs: learner actor requires actor_id,language_profile_id,language_id with composite FK to language_profiles. Negative foreign-user/cross-language evidence fixtures are release-blocking.

Canonical pronunciation state and asynchronous speech-job contract

State	Meaning	Completion / evidence / Auto-advance
speech_correct	Usable, sufficiently confident target-feature match	Positive bounded Pronunciation evidence. Auto-advance allowed after feedback dwell if setting ON; otherwise show “Далі”.
speech_incorrect	Usable, sufficiently confident mismatch	Bounded negative/practice-needed evidence only for targeted pronunciation facet. Non-blocking. Auto-advance may proceed after feedback dwell if setting ON; otherwise show Retry + Continue. Never counts as success.
speech_unavailable	Mic denied/absent, timeout, low confidence, provider failure	No mastery update. Non-blocking. Auto-advance may proceed after neutral status dwell if setting ON; otherwise Retry + Continue.

Canonical state names are speech_correct, speech_incorrect and speech_unavailable. Legacy labels speech_feedback_correct and speech_feedback_retry are invalid.

speech_job lifecycle

created -> uploaded -> queued -> processing -> succeeded | unavailable | failed | cancelled | expired. Network retries reuse the same attempt_id/idempotency key and existing job; internal provider retries create no learner attempt. succeeded settles exactly once as speech_correct/speech_incorrect; unavailable settles exactly once as speech_unavailable; failed/expired settle as speech_unavailable after retry policy is exhausted. cancelled settles the associated attempt exactly once during /cancel as speech_unavailable(reason=user_cancelled); Continue only advances LessonPlayer after that terminal outcome and creates no new attempt or evidence. Learner Retry after any terminal learner-facing result creates a NEW attempt_id and speech_job. Late/duplicate callbacks are signature-validated, checked against final settlement, logged and ignored after settlement. Client polls GET /api/v1/speech/jobs/{job_id}; SSE MAY be added later. Temporary raw audio is encrypted/transient and deleted immediately after terminal settlement or TTL (maximum 15 minutes by default).

Canonical AI API and persistence mapping

API	Purpose
POST /api/v1/ai/explain	Grounded explanation
POST /api/v1/ai/hint	Bounded hint
POST /api/v1/ai/practice	Extra practice
POST /api/v1/ai/writing-feedback	Writing feedback
POST /api/v1/ai/conversations	Create bounded conversation session
POST /api/v1/ai/conversations/{session_id}/messages	Conversation turn
DELETE /api/v1/ai/conversations/{session_id}	Delete/close learner AI session

Logical concept	Physical entity/table
AIProviderProfile	ai_provider_profiles
AIModelProfile	ai_model_profiles
AIPolicy / permissions / budget eligibility	ai_policies
AIPromptTemplate	ai_prompt_templates
AIRequestRecord / AIExecution / AIRun	ai_runs
AISession	ai_sessions
AIMessage	ai_messages
AIArtifact	ai_artifacts
AIUsage / budget consumption	ai_usage_buckets + audit_events
AIToolAction	ai_tool_actions

Learner AI is OFF by default. Effective enablement order: global AI flag -> capability flag -> approved provider/model/data policy -> learner opt-in and required consent -> quota/budget. Any failed check returns deterministic fallback; no paid fallback is implicit.

Canonical monorepo package map

/apps/web /apps/admin /apps/api /apps/worker

/packages/domain /packages/application /packages/activity-schemas /packages/lesson-engine /packages/mastery-review /packages/ai-core /packages/provider-adapters /packages/api-client /packages/ui /packages/content-tools

/packages/provider-adapters/{ai,speech,tts,image,email,storage}/...

/content/english /content/czech /assets/specs /prisma /infra /docs/adr

There is no separate /packages/ai-adapters in the canonical baseline; AI adapters are a sub-area of /packages/provider-adapters. Provider SDK types remain inside adapter subpackages.

Canonical execution authority and milestone map

Milestone	Authority
P0	Governance freeze, exact baseline, decision log, full MUST trace matrix
P1	Monorepo/bootstrap/CI/config
P2	Data, UUID/stable_code/version_code, language isolation, publication
P3	Lesson Engine + attempts + deterministic state
P4	Two-level Mastery + Review + mistakes + collections
P5	Media/audio/speech + async speech jobs
P6	English E-A1-M01
P7	Czech C-A1-M01 + bilingual isolation proof
P8	Admin/CMS + publication + Admin AI Assist foundation
P9	Optional learner AI + AI Auto allowlist
P10	Onboarding/Home/Explore/Course/Progress/Profile/Settings/ notifications + persistence
P11	Reference hardening + 32 scenarios + technical gates; freeze reference baseline
P12	A1 M02-M12 controlled rollout + pilot/calibration + optional A1 Level Test
P13	Full-system hardening/launch gate: security, privacy, performance, restore/DR, release provenance, legal/product readiness

Only Lingvessa_Implementation_Plan v1.4 uses P0-P13 numbering. Any older P-number labels in other specifications are descriptive history and MUST be mapped to these milestones rather than treated as execution authority.

P10 product surfaces and settings acceptance

- Onboarding: choose real English or Czech language profile; language switch preserves independent state.
- Home: current course, due review, progress/streak summary and resume entry; no AI dependency.
- Explore/Course: level/module/lesson navigation, prerequisite warning without hard level lock.
- Progress: six global skill projections, course/level progress and assessment-only reading subscore when a Level Test exists.
- Profile/Settings: target language, en-GB/en-US, UI locale, Auto-advance, audio/speech preferences, learner AI opt-in, notifications, reset lesson/module/all progress, delete account.
- Notifications/quiet hours: timezone-aware IANA settings, quiet-hours persistence and DST tests; notification delivery can be disabled without affecting learning.
- All settings survive reload/login and are server-authoritative with offline/client cache reconciliation.

Appendix D. Full MUST requirement trace matrix

Source	Requirement	Milestone(s)	Owner	Verification
Formal	ROLE-001	P1/P8/P13	Product + relevant engineering owner	Negative permission matrix.
Formal	ROLE-002	P1/P8/P13	Product + relevant engineering owner	actor/action/target/ time/result; no secrets.
Formal	ROLE-003	P1/P8/P13	Product + relevant engineering owner	Workflow/API denies transition.
Formal	ROLE-004	P1/P8/P13	Product + relevant engineering owner	Unknown action denied by policy engine.
Formal	LANG-001	P2/P7/P13	Product + relevant engineering owner	Both open real catalog/lesson flow.
Formal	LANG-002	P2/P7/P13	Product + relevant engineering owner	Switch leaves other profile unchanged.
Formal	LANG-003	P2/P7/P13	Product + relevant	Validator rejects missing

			engineering owner	scope where applicable.
Formal	LANG-004	P2/P7/P13	Product + relevant engineering owner	Script lint fixtures.
Formal	LANG-005	P2/P7/P13	Product + relevant engineering owner	Locale switch does not change target content.
Formal	PED-001	P10/P13	Product + relevant engineering owner	outcome_refs + final task trace.
Formal	PED-002	P10/P13	Product + relevant engineering owner	Scaffolded path review.
Formal	PED-003	P10/P13	Product + relevant engineering owner	Near-duplicate policy.
Formal	PED-004	P10/P13	Product + relevant engineering owner	Coverage/evidence report; no official certificate claim.
Formal	CURR-001	P10/P13	Product + relevant engineering owner	EN/CS fixtures without engine fork.
Formal	CURR-002	P10/P13	Product + relevant engineering owner	Publish validation rejects missing map.
Formal	CURR-003	P10/P13	Product + relevant engineering owner	Admin can inspect prerequisite path.
Formal	CURR-004	P10/P13	Product + relevant engineering owner	UI warns gaps, allows access.
Formal	LESS-001	P3/P11	Product + relevant engineering owner	English/Czech contract fixtures.
Formal	LESS-002	P3/P11	Product + relevant engineering owner	Case/whitespace/ punctuation/diacritics/ variants fixtures.
Formal	LESS-003	P3/P11	Product + relevant engineering owner	Automated + linguistic review.
Formal	LESS-005	P3/P11	Product + relevant engineering owner	Keyboard E2E core types.
Formal	LESS-006	P3/P11	Product + relevant engineering owner	Renderer error fixture.
Formal	STATE-001	P3/P11	Product + relevant engineering owner	Duplicate attempt_id no duplicate score/effects.
Formal	STATE-002	P3/P11	Product + relevant engineering owner	assisted evidence class.
Formal	STATE-003	P3/P11	Product + relevant engineering owner	E2E mistake loop.
Formal	STATE-004	P3/P11	Product + relevant engineering owner	Error tag/cause review.
Formal	MAST-001	P4/P11	Product + relevant engineering owner	Data/API contract.
Formal	MAST-002	P4/P11	Product + relevant engineering owner	Deterministic tests.
Formal	MAST-003	P4/P11	Product + relevant engineering owner	No duplicate mastery rows.
Formal	MAST-004	P4/P11	Product + relevant engineering owner	Versioned recompute.
Formal	MAST-005	P4/P11	Product + relevant engineering owner	Reset/transition preserves evidence unless explicit.
Formal	MAST-006	P4/P11	Product + relevant engineering owner	Schema + projection tests.
Formal	MAST-007	P4/P11	Product + relevant engineering owner	AI/offline comparison produces same core mastery.
Formal	REV-001	P4/P11	Product + relevant engineering owner	No cross-language item.
Formal	REV-002	P4/P11	Product + relevant engineering owner	Scheduler unit tests.
Formal	REV-003	P4/P11	Product + relevant engineering owner	Knowledge refs + normalized hash.
Formal	AUD-001	P5/P11	Product + relevant engineering owner	Playback/duration/ language QA.
Formal	AUD-002	P5/P11	Product + relevant engineering owner	Fresh browser E2E.
Formal	AUD-003	P5/P11	Product + relevant engineering owner	Storage/log audit.
Formal	AUD-004	P5/P11	Product + relevant engineering owner	EN/US/CS fixtures.
Formal	AUD-005	P5/P11	Product + relevant engineering owner	Accessibility review.
Formal	AUD-007	P5/P11	Product + relevant engineering owner	Speech state E2E + mastery assertion.
Formal	COLL-001	P4/P11	Product + relevant engineering owner	API ownership/language tests.
Formal	COLL-002	P4/P11	Product + relevant	Referential integrity.

			engineering owner	
Formal	CONT-001	P2/P7/P13	Product + relevant engineering owner	Edit creates new version.
Formal	CONT-002	P2/P7/P13	Product + relevant engineering owner	Staging rollback.
Formal	CONT-003	P2/P7/P13	Product + relevant engineering owner	Workflow negative tests.
Formal	CONT-004	P2/P7/P13	Product + relevant engineering owner	Impact mapping report.
Formal	MED-001	P10/P13	Product + relevant engineering owner	Published lesson complete with system assets.
Formal	MED-002	P10/P13	Product + relevant engineering owner	Preflight existence/checksum/format/CDN.
Formal	MED-003	P10/P13	Product + relevant engineering owner	Series assets share profile IDs.
Formal	MED-004	P10/P13	Product + relevant engineering owner	Platform-controlled storage after ingest.
Formal	MED-005	P10/P13	Product + relevant engineering owner	Rollback cache-safe.
Formal	MED-006	P10/P13	Product + relevant engineering owner	Media gate blocks unknown identity provenance.
Formal	AI-001	P8/P9/P13	Product + relevant engineering owner	Provider-offline EN/CS reference E2E.
Formal	AI-002	P8/P9/P13	Product + relevant engineering owner	AI cannot directly set correct/mastery/published/role.
Formal	AI-003	P8/P9/P13	Product + relevant engineering owner	Audit/support reproduction.
Formal	AI-004	P8/P9/P13	Product + relevant engineering owner	Cost guard tests.
Formal	AI-005	P8/P9/P13	Product + relevant engineering owner	Permission matrix.
Formal	AI-006	P8/P9/P13	Product + relevant engineering owner	Authorization negative tests.
Formal	AI-007	P8/P9/P13	Product + relevant engineering owner	Grounding/eval fixtures.
Formal	AI-008	P8/P9/P13	Product + relevant engineering owner	No direct provider asset binding in published lesson.
Formal	UX-001	P10/P13	Product + relevant engineering owner	No accidental cross-language lesson.
Formal	UX-002	P10/P13	Product + relevant engineering owner	Breakpoint QA.
Formal	UX-003	P10/P13	Product + relevant engineering owner	E2E correct/incorrect/reveal/network/speech.
Formal	UX-004	P10/P13	Product + relevant engineering owner	Toggle/state test.
Formal	UX-005	P10/P13	Product + relevant engineering owner	Scope shown before mutation.
Formal	A11Y-001	P10/P13	Product + relevant engineering owner	Automated + manual evidence.
Formal	A11Y-002	P10/P13	Product + relevant engineering owner	Desktop/mobile SR scenarios.
Formal	A11Y-003	P10/P13	Product + relevant engineering owner	Text/icon state remains clear.
Formal	A11Y-004	P10/P13	Product + relevant engineering owner	DOM/lang audit.
Formal	SET-001	P10/P13	Product + relevant engineering owner	Restore on another device.
Formal	SET-002	P10/P13	Product + relevant engineering owner	DST/off tests.
Formal	SET-003	P10/P13	Product + relevant engineering owner	Deletion job/audit.
Formal	SET-004	P10/P13	Product + relevant engineering owner	Provider-disabled E2E.
Formal	TECH-001	P10/P13	Product + relevant engineering owner	Production uses schemas/tests/persistence.
Formal	TECH-002	P10/P13	Product + relevant engineering owner	Architecture lint.
Formal	TECH-003	P10/P13	Product + relevant engineering owner	CI reproducible build.
Formal	TECH-004	P10/P13	Product + relevant engineering owner	Secret scan + bundle inspection.

Formal	DATA-001	P2/P7/P13	Product + relevant engineering owner	Rename/reorder keeps refs/progress.
Formal	DATA-002	P2/P7/P13	Product + relevant engineering owner	No generic update API.
Formal	DATA-003	P2/P7/P13	Product + relevant engineering owner	IDOR tests.
Formal	DATA-004	P2/P7/P13	Product + relevant engineering owner	Old fixtures render/migrate.
Formal	DATA-005	P2/P7/P13	Product + relevant engineering owner	Outage does not affect completion.
Formal	DATA-006	P2/P7/P13	Product + relevant engineering owner	Schema prevents dimension collapse.
Formal	API-001	P10/P13	Product + relevant engineering owner	UI distinguishes validation/auth/conflict/rate/service unavailable.
Formal	API-002	P10/P13	Product + relevant engineering owner	Retry tests.
Formal	API-003	P10/P13	Product + relevant engineering owner	Tampering has no effect.
Formal	API-004	P10/P13	Product + relevant engineering owner	Failed/stuck visible; no infinite retry.
Formal	API-006	P10/P13	Product + relevant engineering owner	Authorization/evidence tests.
Formal	SEC-001	P1/P8/P13	Product + relevant engineering owner	ASVS-oriented IDOR/privilege tests.
Formal	SEC-002	P1/P8/P13	Product + relevant engineering owner	Automated canary scan.
Formal	SEC-003	P1/P8/P13	Product + relevant engineering owner	E2E lifecycle.
Formal	SEC-004	P1/P8/P13	Product + relevant engineering owner	No MFA -> no privileged command.
Formal	SEC-005	P1/P8/P13	Product + relevant engineering owner	Release gate artifacts.
Formal	SEC-006	P1/P8/P13	Product + relevant engineering owner	Registration gating.
Formal	SEC-007	P1/P8/P13	Product + relevant engineering owner	Privacy route tests.
Formal	LEGAL-001	P10/P13	Product + relevant engineering owner	Unknown/restricted blocks publish.
Formal	LEGAL-002	P10/P13	Product + relevant engineering owner	Documented audit before public launch.
Formal	LEGAL-003	P10/P13	Product + relevant engineering owner	Decision log.
Formal	LEGAL-004	P10/P13	Product + relevant engineering owner	Vendor register version/date/purpose.
Formal	NFR-REL-001	P10/P13	Product + relevant engineering owner	Fault injection.
Formal	NFR-DR-001	P10/P13	Product + relevant engineering owner	Restore drill.
Formal	NFR-SCALE-001	P10/P13	Product + relevant engineering owner	Multi-instance test.
Formal	NFR-LOC-001	P10/P13	Product + relevant engineering owner	Normalization fixtures.
Formal	NFR-AI-001	P8/P9/P13	Product + relevant engineering owner	Provider-offline SLO/fault test.
Formal	ACC-001	P10/P13	Product + relevant engineering owner	New-account E2E both languages.
Formal	ACC-002	P10/P13	Product + relevant engineering owner	Mixed data assertions.
Formal	ACC-003	P10/P13	Product + relevant engineering owner	Fixture swap without code edit.
Formal	ACC-004	P10/P13	Product + relevant engineering owner	Manifest preflight/provider offline.
Formal	ACC-005	P10/P13	Product + relevant engineering owner	Deterministic state/mastery/scheduler tests.
Formal	ACC-006	P10/P13	Product + relevant engineering owner	Visual/E2E matrix.
Formal	ACC-007	P10/P13	Product + relevant engineering owner	Automated + manual WCAG evidence.
Formal	ACC-008	P10/P13	Product + relevant engineering owner	Documented drill.
Formal	ACC-009	P10/P13	Product + relevant engineering owner	Threat model + lifecycle checklist.
Formal	ACC-010	P10/P13	Product + relevant engineering owner	Review record.
Formal	ACC-011	P10/P13	Product + relevant	Fault injection.

			engineering owner	
Formal	ACC-012	P10/P13	Product + relevant engineering owner	Restore report.
Formal	ACC-013	P10/P13	Product + relevant engineering owner	Speech fixtures and mastery assertions.
Formal	ACC-014	P10/P13	Product + relevant engineering owner	Schema/projection/ review tests.
Formal	ACC-015	P10/P13	Product + relevant engineering owner	Provider-offline E2E + budget guard.
Technical Design	TD-GOV-001	P1-P11	Backend/platform + QA	Release checklist has no orphan MUST requirement.
Technical Design	TD-GOV-002	P1-P11	Backend/platform + QA	All providers disabled still passes core English/Czech reference E2E.
Technical Design	TD-GOV-003	P1-P11	Backend/platform + QA	Change log, migration/compatibility note and affected tests exist.
Technical Design	TD-GOV-004	P1-P11	Backend/platform + QA	Provider replacement contract tests pass; no provider URL in canonical content.
Technical Design	TD-MOD-001	P1-P11	Backend/platform + QA	Dependency lint passes.
Technical Design	TD-MOD-002	P1-P11	Backend/platform + QA	Architecture tests + code review.
Technical Design	TD-MOD-003	P1-P11	Backend/platform + QA	Forbidden-import rule blocks PR.
Technical Design	TD-MOD-004	P1-P11	Backend/platform + QA	Red-team tool call cannot bypass RBAC/object policy.
Technical Design	TD-ENV-001	P1-P11	Backend/platform + QA	Startup/config contract tests.
Technical Design	TD-ENV-002	P1-P11	Backend/platform + QA	Secret scan + bundle/log inspection.
Technical Design	TD-ENV-003	P1-P11	Backend/platform + QA	Container + externalized config portability review.
Technical Design	TD-ENV-004	P1-P11	Backend/platform + QA	Config test rejects billable route before provider call.
Technical Design	TD-DATA-001	P2/P7/P13	Backend/platform + QA	Negative DB/service fixtures.
Technical Design	TD-DATA-002	P2/P7/P13	Backend/platform + QA	Concurrent duplicate tests.
Technical Design	TD-DATA-003	P2/P7/P13	Backend/platform + QA	Schema/privacy audit.
Technical Design	TD-DATA-004	P2/P7/P13	Backend/platform + QA	Sink outage does not affect lesson commit.
Technical Design	TD-CONT-001	P2/P7/P13	Backend/platform + QA	Checksum/DB tests block in-place overwrite.
Technical Design	TD-CONT-002	P2/P7/P13	Backend/platform + QA	Mutation-after-approval workflow test.
Technical Design	TD-CONT-003	P2/P7/P13	Backend/platform + QA	Service role/policy rejects direct publish.
Technical Design	TD-CONT-004	P2/P7/P13	Backend/platform + QA	No silent skill/facet mastery loss.
Technical Design	TD-AIDATA-001	P8/P9/P13	Backend/platform + QA	Schema/log review finds no reasoning field.
Technical Design	TD-AIDATA-002	P8/P9/P13	Backend/platform + QA	Routing negative test.
Technical Design	TD-AIDATA-003	P8/P9/P13	Backend/platform + QA	Zero-budget test makes no billable request.
Technical Design	TD-AIDATA-004	P8/P9/P13	Backend/platform + QA	IDOR + deletion lifecycle tests.
Technical Design	TD-ACT-001	P3/P11	Backend/platform + QA	Every published type has English+Czech fixture.
Technical Design	TD-ACT-002	P3/P11	Backend/platform + QA	Provider disable does not break renderer.
Technical Design	TD-ACT-003	P3/P11	Backend/platform + QA	E2E invalid fixture.
Technical Design	TD-STATE-001	P3/P11	Backend/platform + QA	State/property tests.
Technical Design	TD-STATE-002	P3/P11	Backend/platform + QA	Fault matrix.
Technical Design	TD-STATE-003	P3/P11	Backend/platform + QA	Tampered/AI response tests.
Technical Design	TD-STATE-004	P3/P11	Backend/platform + QA	A11y motion test.
Technical Design	TD-ATT-001	P3/P11	Backend/platform + QA	Concurrent retry test creates one effect set.
Technical Design	TD-ATT-002	P3/P11	Backend/platform + QA	Injected failure leaves

				all-or-none.
Technical Design	TD-ATT-003	P3/P11	Backend/platform + QA	Historical replay explainable.
Technical Design	TD-ATT-004	P3/P11	Backend/platform + QA	Trace review + timeout fixture.
Technical Design	TD-MAST-001	P4/P11	Backend/platform + QA	Recompute fixtures.
Technical Design	TD-MAST-002	P4/P11	Backend/platform + QA	Duplicate surface test.
Technical Design	TD-MAST-003	P4/P11	Backend/platform + QA	Reset/transition never deletes facet evidence accidentally.
Technical Design	TD-MAST-004	P4/P11	Backend/platform + QA	AI-help then answer test.
Technical Design	TD-MAST-005	P4/P11	Backend/platform + QA	AI evidence permission test.
Technical Design	TD-REV-001	P4/P11	Backend/platform + QA	Mixed dataset test.
Technical Design	TD-REV-002	P4/P11	Backend/platform + QA	Golden tests.
Technical Design	TD-REV-003	P4/P11	Backend/platform + QA	Practice integration test.
Technical Design	TD-AILRN-001	P8/P9/P13	Backend/platform + QA	Contract tests for English+Czech.
Technical Design	TD-AILRN-002	P8/P9/P13	Backend/platform + QA	Adversarial prompt/hint test.
Technical Design	TD-AILRN-003	P8/P9/P13	Backend/platform + QA	Invalid output retry/fallback.
Technical Design	TD-AILRN-004	P8/P9/P13	Backend/platform + QA	Quota/outage/deletion tests.
Technical Design	TD-AILRN-005	P8/P9/P13	Backend/platform + QA	Mastery protection test.
Technical Design	TD-AI-001	P8/P9/P13	Backend/platform + QA	Sensitive-route negative tests.
Technical Design	TD-AI-002	P8/P9/P13	Backend/platform + QA	Mock billing route sees zero network calls.
Technical Design	TD-AI-003	P8/P9/P13	Backend/platform + QA	Fault/incident drill.
Technical Design	TD-AI-004	P8/P9/P13	Backend/platform + QA	Invalid JSON/schema fixtures.
Technical Design	TD-AI-005	P8/P9/P13	Backend/platform + QA	Adapter substitution compile/contract test.
Technical Design	TD-AIADM-001	P8/P9/P13	Backend/platform + QA	Negative permission matrix.
Technical Design	TD-AIADM-002	P8/P9/P13	Backend/platform + QA	Prompt injection/tool call cannot execute them.
Technical Design	TD-AIADM-003	P8/P9/P13	Backend/platform + QA	Admin workflow audit fixture.
Technical Design	TD-AIADM-004	P8/P9/P13	Backend/platform + QA	RBAC/MFA setting test.
Technical Design	TD-JOB-001	P1-P11	Backend/platform + QA	Crash/double-delivery tests.
Technical Design	TD-JOB-002	P1-P11	Backend/platform + QA	Forged/replay callbacks rejected.
Technical Design	TD-JOB-003	P1-P11	Backend/platform + QA	Permanent error reaches terminal state.
Technical Design	TD-JOB-004	P1-P11	Backend/platform + QA	Fault injection alerts.
Technical Design	TD-API-001	P1-P11	Backend/platform + QA	Tampering matrix.
Technical Design	TD-API-002	P1-P11	Backend/platform + QA	Provider mock sees no blocked request.
Technical Design	TD-API-003	P1-P11	Backend/platform + QA	IDOR/RBAC E2E.
Technical Design	TD-API-004	P1-P11	Backend/platform + QA	Contract diff gate.
Technical Design	TD-CMS-001	P1-P11	Backend/platform + QA	Save targets draft version only.
Technical Design	TD-CMS-002	P1-P11	Backend/platform + QA	Renderer contract test.
Technical Design	TD-CMS-003	P1-P11	Backend/platform + QA	Negative privileged action test.
Technical Design	TD-CMS-004	P1-P11	Backend/platform + QA	RBAC/MFA test.
Technical Design	TD-SEC-001	P1/P8/P13	Backend/platform + QA	Approved control matrix before public launch.
Technical Design	TD-SEC-002	P1/P8/P13	Backend/platform + QA	Red-team injection test.
Technical Design	TD-SEC-003	P1/P8/P13	Backend/platform + QA	Malformed/polyglot/ invalid AI output fixtures.
Technical Design	TD-SEC-004	P1/P8/P13	Backend/platform + QA	Tabletop/fault drill.
Technical Design	TD-REL-001	P1-P11	Backend/platform + QA	Fault-injection one durable effect.
Technical Design	TD-REL-002	P1-P11	Backend/platform + QA	Restore drill.
Technical Design	TD-REL-003	P1-P11	Backend/platform + QA	Circuit-break/load test.
Technical Design	TD-REL-004	P1-P11	Backend/platform + QA	Cost fault test.
Technical Design	TD-OBS-001	P1-P11	Backend/platform + QA	End-to-end staging trace.

Technical Design	TD-OBS-002	P1-P11	Backend/platform + QA	Usage dashboard test.
Technical Design	TD-OBS-003	P1-P11	Backend/platform + QA	Canary redaction tests.
Technical Design	TD-OBS-004	P1-P11	Backend/platform + QA	Owner/date/drill evidence.
Technical Design	TD-CICD-001	P1-P11	Backend/platform + QA	Release record reconstructs deployed state.
Technical Design	TD-CICD-002	P1-P11	Backend/platform + QA	Staging eval + flag rollback.
Technical Design	TD-CICD-003	P1-P11	Backend/platform + QA	No billable mock calls.
Technical Design	TD-CICD-004	P1-P11	Backend/platform + QA	CI report.
Curriculum	CUR-A1-001	P6/P7/P12	Curriculum + language lead	Separate curriculum/profile/progress/mastery/review.
Curriculum	CUR-A1-002	P6/P7/P12	Curriculum + language lead	Manifest totals 60 units per language.
Curriculum	CUR-A1-003	P6/P7/P12	Curriculum + language lead	Coverage report no unmapped module.
Curriculum	CUR-A1-004	P6/P7/P12	Curriculum + language lead	Context, active retrieval, visible result.
Curriculum	CUR-A1-005	P6/P7/P12	Curriculum + language lead	No unresolved reference.
Curriculum	CUR-A1-006	P6/P7/P12	Curriculum + language lead	Lexical/grammar item metadata complete.
Curriculum	CUR-A1-007	P6/P7/P12	Curriculum + language lead	Evidence distinguishes e.g. meaning vs spelling vs production.
Curriculum	CUR-A1-008	P6/P7/P12	Curriculum + language lead	Cross-language isolation test.
Curriculum	CUR-A1-009	P6/P7/P12	Curriculum + language lead	Attempt/evidence/scheduling test.
Curriculum	CUR-A1-010	P6/P7/P12	Curriculum + language lead	Correct/mismatch/denial/low-confidence/provider-failure paths.
Curriculum	CUR-A1-011	P6/P7/P12	Curriculum + language lead	en-US variants/voices/notes configured where meaningful.
Curriculum	CUR-A1-012	P6/P7/P12	Curriculum + language lead	M01 + later morphology scenario.
Curriculum	CUR-A1-013	P6/P7/P12	Curriculum + language lead	Coverage all four lessons + recycle links.
Curriculum	CUR-A1-014	P6/P7/P12	Curriculum + language lead	No invented threshold in config.
Curriculum	CUR-A1-015	P6/P7/P12	Curriculum + language lead	asset IDs/transcripts/rights resolve.
Curriculum	CUR-A1-016	P6/P7/P12	Curriculum + language lead	Originality/provenance review.
Curriculum	CUR-A1-017	P6/P7/P12	Curriculum + language lead	Keyboard/touch/transcript/non-mic QA.
Curriculum	CUR-A1-018	P6/P7/P12	Curriculum + language lead	Audit explains evidence effect.
Curriculum	CUR-A1-019	P6/P7/P12	Curriculum + language lead	AI off/outage leaves full core path; grounding fixture valid.
Curriculum	CUR-A1-020	P6/P7/P12	Curriculum + language lead	Evidence API rejects unvalidated AI authority.
Curriculum	CUR-A1-021	P6/P7/P12	Curriculum + language lead	Practice-only default test.
Curriculum	CUR-A1-022	P6/P7/P12	Curriculum + language lead	EN/CS reference validation suite passes.
Reference	REF-A1-001	P6/P7/P11	Reference implementation + QA	Schema validation passes without language-specific engine fork.
Reference	REF-A1-002	P6/P7/P11	Reference implementation + QA	English/Czech switching preserves independent state.
Reference	REF-A1-003	P6/P7/P11	Reference implementation + QA	Manifest validator finds all required stages.
Reference	REF-A1-004	P6/P7/P11	Reference implementation + QA	Next/auto-advance state tests pass.
Reference	REF-A1-005	P6/P7/P11	Reference implementation + QA	Evidence and scheduler events contain reason/rule version.
Reference	REF-A1-006	P6/P7/P11	Reference implementation + QA	Correct, mismatch, denial, low-confidence and provider-failure

				paths complete safely.
Reference	REF-A1-007	P6/P7/P11	Reference implementation + QA	Meaning/spelling/ listening/pronunciation/ register evidence remains distinguishable.
Reference	REF-A1-008	P6/P7/P11	Reference implementation + QA	Selected variety controls voices, notes and accepted variants.
Reference	REF-A1-009	P6/P7/P11	Reference implementation + QA	Strict/form-focused and meaning-focused tests produce different facet evidence.
Reference	REF-A1-010	P6/P7/P11	Reference implementation + QA	asset_id + asset_version + QA + provenance/rights all resolve.
Reference	REF-A1-011	P6/P7/P11	Reference implementation + QA	Sample profiles and system media cover every required path.
Reference	REF-A1-012	P6/P7/P11	Reference implementation + QA	Schema/evaluator/UI/AI-adapter changes run the regression suite.
Reference	REF-A1-013	P6/P7/P11	Reference implementation + QA	AI disabled/outage/quota tests preserve complete non-AI lesson path.
Reference	REF-A1-014	P6/P7/P11	Reference implementation + QA	Only approved evaluator/rubric/workflo w transitions can create bounded evidence/publish change.
AI	AI-GOV-001	P8/P9/P13	AI/application + security	Provider-offline E2E completes both English and Czech reference modules excluding explicitly optional AI activities.
AI	AI-GOV-002	P8/P9/P13	AI/application + security	No AI endpoint can directly set published=true, correct=true, mastery estimate or role.
AI	AI-GOV-003	P8/P9/P13	AI/application + security	Audit/support can reproduce configuration of a representative AI result.
AI	AI-GOV-004	P8/P9/P13	AI/application + security	Domain and content schemas contain no provider SDK types or canonical provider URLs.
AI	AI-FALL-001	P8/P9/P13	AI/application + security	Fault matrix for timeout/429/5xx/provide r disabled produces no dead end.
AI	AI-FALL-002	P8/P9/P13	AI/application + security	Injected AI failure after attempt commit leaves progress unchanged and durable.
AI	AI-FALL-003	P8/P9/P13	AI/application + security	Sensitive request is rejected rather than routed to unapproved provider.
AI	AI-PROV-001	P8/P9/P13	AI/application + security	Swap adapter without changing learner/content schema.
AI	AI-PROV-002	P8/P9/P13	AI/application + security	Sensitive fixture never reaches cheaper but unapproved provider.
AI	AI-PROV-003	P8/P9/P13	AI/application + security	Provider registry records verified_at, terms/data-policy refs and owner.
AI	AI-PROV-004	P8/P9/P13	AI/application + security	Disable primary model during E2E and complete non-AI flow.
AI	AI-COST-001	P8/P9/P13	AI/application + security	Integration test with paid candidate selected returns POLICY_BLOCKED before network call.
AI	AI-COST-002	P8/P9/P13	AI/application + security	Usage dashboard reconciles provider response headers/usage

				fields for sampled calls.
AI	AI-COST-003	P8/P9/P13	AI/application + security	Admin can preview quota impact and abort before provider call.
AI	AI-LRN-001	P8/P9/P13	AI/application + security	Response without valid refs is rejected or downgraded to generic non-authoritative assistance.
AI	AI-LRN-002	P8/P9/P13	AI/application + security	Golden bilingual eval suite passes approved rule cases.
AI	AI-GRD-001	P8/P9/P13	AI/application + security	Audit result identifies source IDs/versions.
AI	AI-GRD-002	P8/P9/P13	AI/application + security	Privacy test inspects outbound payload allowlist.
AI	AI-GRD-003	P8/P9/P13	AI/application + security	Broken grounding fixture produces safe refusal or curated fallback.
AI	AI-MAST-001	P8/P9/P13	AI/application + security	Attempt/evidence API rejects AI confidence as authority.
AI	AI-MAST-002	P8/P9/P13	AI/application + security	Evidence fixtures distinguish meaning vs spelling vs production.
AI	AI-MAST-003	P8/P9/P13	AI/application + security	Level Test config has deterministic/rubric evidence and calibration record.
AI	AI-ADM-001	P8/P9/P13	AI/application + security	Unknown action denied by policy engine.
AI	AI-ADM-002	P8/P9/P13	AI/application + security	Authorization matrix tests fail direct calls.
AI	AI-ADM-003	P8/P9/P13	AI/application + security	Admin audit opens complete change trail.
AI	AI-PRIV-001	P1/P8/P13	AI/application + security	Privacy fixture detects no unrelated profile, tokens or raw audio.
AI	AI-PRIV-002	P1/P8/P13	AI/application + security	Routing blocks provider with unknown policy for learner personal data.
AI	AI-PRIV-003	P1/P8/P13	AI/application + security	Deletion drill reports completed/failed downstream steps.
AI	AI-SEC-001	P1/P8/P13	AI/application + security	Red-team prompt cannot publish/change role/delete.
AI	AI-SEC-002	P1/P8/P13	AI/application + security	Invalid/unauthorized tool call rejected and audited.
AI	AI-SEC-003	P1/P8/P13	AI/application + security	Canary secret/raw learner text absent from telemetry sink.
AI	AI-PRM-001	P8/P9/P13	AI/application + security	CI/staging report links prompt version to pass results.
AI	AI-PRM-002	P8/P9/P13	AI/application + security	Secret/privacy lint passes.
AI	AI-PRM-003	P8/P9/P13	AI/application + security	Router cannot promote untested model to learner default.
AI	AI-OBS-001	P8/P9/P13	AI/application + security	Support can trace request without raw secret content.
AI	AI-OBS-002	P8/P9/P13	AI/application + security	Dashboard and alert show zero paid calls under start policy.
AI	AI-OBS-003	P8/P9/P13	AI/application + security	Incident drill disables AI and core lesson remains usable.
Technical	TS-GOV-001	P1-P13	Engineering + QA	Release checklist не містить orphan MUST.
Technical	TS-GOV-002	P1-P13	Engineering + QA	ADR + spec update.
Technical	TS-GOV-003	P1-P13	Engineering + QA	No hidden production default.
Technical	TS-GOV-004	P1-P13	Engineering + QA	Change log + migration note.
Technical	TS-ARCH-001	P1-P13	Engineering + QA	No language-specific

				engine fork.
Technical	TS-ARCH-002	P1-P13	Engineering + QA	AI/provider outage fault test.
Technical	TS-ARCH-003	P1-P13	Engineering + QA	Cache/AI reset does not alter progress.
Technical	TS-ARCH-004	P1-P13	Engineering + QA	Architecture lint.
Technical	TS-TECH-001	P1-P13	Engineering + QA	CI install/build.
Technical	TS-TECH-002	P1-P13	Engineering + QA	Lifecycle job.
Technical	TS-TECH-003	P1-P13	Engineering + QA	Patch cadence.
Technical	TS-TECH-004	P1-P13	Engineering + QA	Upgrade PR gate.
Technical	TS-TECH-005	P1-P13	Engineering + QA	Forbidden import / adapter contract.
Technical	TS-MOD-001	P1-P13	Engineering + QA	Dependency lint.
Technical	TS-MOD-002	P1-P13	Engineering + QA	Forbidden import rule.
Technical	TS-MOD-003	P1-P13	Engineering + QA	Architecture/security tests.
Technical	TS-MOD-004	P1-P13	Engineering + QA	Integration tests.
Technical	TS-ENV-001	P1-P13	Engineering + QA	Startup config tests.
Technical	TS-ENV-002	P1-P13	Engineering + QA	Secret scans.
Technical	TS-ENV-003	P1-P13	Engineering + QA	Release record.
Technical	TS-ENV-004	P1-P13	Engineering + QA	Config test + audit.
Technical	TS-ENV-005	P1-P13	Engineering + QA	Negative tests.
Technical	TS-REPO-001	P1-P13	Engineering + QA	Clean checkout rebuilds artifacts.
Technical	TS-REPO-002	P1-P13	Engineering + QA	Secret/size checks.
Technical	TS-REPO-003	P1-P13	Engineering + QA	Release reconstruction drill.
Technical	TS-DATA-001	P2/P7/P13	Engineering + QA	Duplicate rejected; switch preserves both states.
Technical	TS-DATA-002	P2/P7/P13	Engineering + QA	Negative mixed-data tests.
Technical	TS-DATA-003	P2/P7/P13	Engineering + QA	IDOR tests.
Technical	TS-DATA-004	P2/P7/P13	Engineering + QA	Checksum fixture.
Technical	TS-DATA-005	P2/P7/P13	Engineering + QA	Concurrent conflict tests.
Technical	TS-DATA-006	P2/P7/P13	Engineering + QA	No generic update endpoint.
Technical	TS-DATA-007	P2/P7/P13	Engineering + QA	Audit/privacy fixture.
Technical	TS-DATA-008	P2/P7/P13	Engineering + QA	Schema review.
Technical	TS-CONT-001	P2/P7/P13	Engineering + QA	Checksum and overwrite tests.
Technical	TS-CONT-002	P2/P7/P13	Engineering + QA	Manifest restore.
Technical	TS-CONT-003	P2/P7/P13	Engineering + QA	Preflight blocks missing mapping.
Technical	TS-CONT-004	P2/P7/P13	Engineering + QA	No silent mastery loss.
Technical	TS-CONT-005	P2/P7/P13	Engineering + QA	Workflow test.
Technical	TS-ACT-001	P3/P11	Engineering + QA	Contract suite.
Technical	TS-ACT-002	P3/P11	Engineering + QA	Fixture swap.
Technical	TS-ACT-003	P3/P11	Engineering + QA	E2E error boundary.
Technical	TS-ACT-004	P3/P11	Engineering + QA	AI-off E2E.
Technical	TS-EVAL-001	P1-P13	Engineering + QA	Tamper test.
Technical	TS-EVAL-002	P1-P13	Engineering + QA	Independent vs assisted test.
Technical	TS-EVAL-003	P1-P13	Engineering + QA	Language fixtures.
Technical	TS-EVAL-004	P1-P13	Engineering + QA	Speech fixtures.
Technical	TS-EVAL-005	P1-P13	Engineering + QA	Provider disagreement test.
Technical	TS-ATT-001	P3/P11	Engineering + QA	Concurrent retry test.
Technical	TS-ATT-002	P3/P11	Engineering + QA	Retry/conflict tests.
Technical	TS-ATT-003	P3/P11	Engineering + QA	Fault injection.
Technical	TS-ATT-004	P3/P11	Engineering + QA	Historical replay.
Technical	TS-ATT-005	P3/P11	Engineering + QA	Consumer failure test.
Technical	TS-MAST-001	P4/P11	Engineering + QA	Recompute fixture.
Technical	TS-MAST-002	P4/P11	Engineering + QA	Duplicate-surface test.
Technical	TS-MAST-003	P4/P11	Engineering + QA	Property tests.
Technical	TS-MAST-004	P4/P11	Engineering + QA	State/scheduler test.
Technical	TS-MAST-005	P4/P11	Engineering + QA	AI evidence gate test.
Technical	TS-REV-001	P4/P11	Engineering + QA	Mixed dataset.
Technical	TS-REV-002	P4/P11	Engineering + QA	Golden tests.
Technical	TS-REV-003	P4/P11	Engineering + QA	Policy config test.
Technical	TS-REV-004	P4/P11	Engineering + QA	AI review test.

Technical	TS-SYNC-001	P3/P11	Engineering + QA	E2E every core state.
Technical	TS-SYNC-002	P3/P11	Engineering + QA	Storage inspection.
Technical	TS-SYNC-003	P3/P11	Engineering + QA	Network flap test.
Technical	TS-SYNC-004	P3/P11	Engineering + QA	Offline AI-off E2E.
Technical	TS-API-001	P1-P13	Engineering + QA	Contract diff.
Technical	TS-API-002	P1-P13	Engineering + QA	Negative tests.
Technical	TS-API-003	P1-P13	Engineering + QA	Security matrix.
Technical	TS-API-004	P1-P13	Engineering + QA	Tampering tests.
Technical	TS-API-005	P1-P13	Engineering + QA	OpenAPI diff gate.
Technical	TS-ADM-001	P1-P13	Engineering + QA	Workflow transition tests.
Technical	TS-ADM-002	P1-P13	Engineering + QA	Mutation invalidates approval.
Technical	TS-ADM-003	P1-P13	Engineering + QA	Permission matrix.
Technical	TS-ADM-004	P1-P13	Engineering + QA	Audit event includes old/new safe values.
Technical	TS-ADM-005	P1-P13	Engineering + QA	Retry idempotency.
Technical	TS-JOB-001	P1-P13	Engineering + QA	Crash delivery test.
Technical	TS-JOB-002	P1-P13	Engineering + QA	Duplicate delivery test.
Technical	TS-JOB-003	P1-P13	Engineering + QA	Permanent error test.
Technical	TS-JOB-004	P1-P13	Engineering + QA	Telemetry schema test.
Technical	TS-JOB-005	P1-P13	Engineering + QA	Forged callback test.
Technical	TS-AI-001	P8/P9/P13	Engineering + QA	Adapter substitution contract.
Technical	TS-AI-002	P8/P9/P13	Engineering + QA	Replay/audit record.
Technical	TS-AI-003	P8/P9/P13	Engineering + QA	Zero-budget test.
Technical	TS-AI-004	P8/P9/P13	Engineering + QA	Negative permission matrix.
Technical	TS-AI-005	P8/P9/P13	Engineering + QA	Malformed/hallucinated output tests.
Technical	TS-AI-006	P8/P9/P13	Engineering + QA	Prompt/policy tests.
Technical	TS-AI-007	P8/P9/P13	Engineering + QA	Privacy payload audit.
Technical	TS-AI-008	P8/P9/P13	Engineering + QA	Fault injection.
Technical	TS-MEDIA-001	P5/P11	Engineering + QA	Manifest preflight.
Technical	TS-MEDIA-002	P5/P11	Engineering + QA	Rollback cache test.
Technical	TS-MEDIA-003	P5/P11	Engineering + QA	Speech E2E storage audit.
Technical	TS-MEDIA-004	P5/P11	Engineering + QA	Mastery fixture.
Technical	TS-MEDIA-005	P5/P11	Engineering + QA	Published lesson provider-offline test.
Technical	TS-EVT-001	P1-P13	Engineering + QA	Schema registry tests.
Technical	TS-EVT-002	P1-P13	Engineering + QA	Support trace.
Technical	TS-EVT-003	P1-P13	Engineering + QA	Payload privacy tests.
Technical	TS-EVT-004	P1-P13	Engineering + QA	Sink outage test.
Technical	TS-WEB-001	P10/P13	Engineering + QA	Language switch E2E.
Technical	TS-WEB-002	P10/P13	Engineering + QA	Bundle scan.
Technical	TS-WEB-003	P10/P13	Engineering + QA	Malformed fixture.
Technical	TS-WEB-004	P10/P13	Engineering + QA	AI timeout UI test.
Technical	TS-LESS-001	P3/P11	Engineering + QA	State-machine tests.
Technical	TS-LESS-002	P3/P11	Engineering + QA	No-dead-end E2E.
Technical	TS-LESS-003	P3/P11	Engineering + QA	Correct/reveal/speech tests.
Technical	TS-LESS-004	P3/P11	Engineering + QA	AI interaction state test.
Technical	TS-A11Y-001	P10/P13	Engineering + QA	A11y gate.
Technical	TS-A11Y-002	P10/P13	Engineering + QA	Contract gate.
Technical	TS-A11Y-003	P10/P13	Engineering + QA	Reduced-motion E2E.
Technical	TS-CMS-001	P1-P13	Engineering + QA	Save behavior.
Technical	TS-CMS-002	P1-P13	Engineering + QA	Renderer contract.
Technical	TS-CMS-003	P1-P13	Engineering + QA	Audit test.
Technical	TS-CMS-004	P1-P13	Engineering + QA	Policy matrix.
Technical	TS-CMS-005	P1-P13	Engineering + QA	Publish/rollback/delete/ role/budget E2E.
Technical	TS-AUTH-001	P1/P8/P13	Engineering + QA	Negative matrix.
Technical	TS-AUTH-002	P1/P8/P13	Engineering + QA	MFA E2E.
Technical	TS-AUTH-003	P1/P8/P13	Engineering + QA	Policy test.
Technical	TS-PRIV-001	P1/P8/P13	Engineering + QA	Approved register.
Technical	TS-PRIV-002	P1/P8/P13	Engineering + QA	Storage/provider cleanup audit.
Technical	TS-PRIV-003	P1/P8/P13	Engineering + QA	Provider payload audit.
Technical	TS-PRIV-004	P1/P8/P13	Engineering + QA	Deletion drill.
Technical	TS-SEC-001	P1/P8/P13	Engineering + QA	Control matrix.

Technical	TS-SEC-002	P1/P8/P13	Engineering + QA	CI gate.
Technical	TS-SEC-003	P1/P8/P13	Engineering + QA	Prompt-injection/action tests.
Technical	TS-SEC-004	P1/P8/P13	Engineering + QA	Malformed/polyglot/ schema tests.
Technical	TS-PERF-001	P1-P13	Engineering + QA	Recorded benchmark.
Technical	TS-PERF-002	P1-P13	Engineering + QA	Trace review.
Technical	TS-PERF-003	P1-P13	Engineering + QA	Limiter load test.
Technical	TS-PERF-004	P1-P13	Engineering + QA	Fault test.
Technical	TS-CICD-001	P1-P13	Engineering + QA	Access review.
Technical	TS-CICD-002	P1-P13	Engineering + QA	Staging rehearsal.
Technical	TS-CICD-003	P1-P13	Engineering + QA	Compatibility tests.
Technical	TS-CICD-004	P1-P13	Engineering + QA	AI release gate.
AI	AIA-01	P8/P9/P13	AI/application + QA/security	English and Czech core lesson/review/progress complete with all AI providers disabled.
AI	AIA-02	P8/P9/P13	AI/application + QA/security	Free quota exhaustion causes no paid call; fallback/unavailable only.
AI	AIA-03	P8/P9/P13	AI/application + QA/security	Sensitive learner text never reaches unapproved provider.
AI	AIA-04	P9/P13	AI/application + QA/security	Czech grounded explanation cites exact approved refs; bad grounding rejected.
AI	AIA-05	P9/P13	AI/application + QA/security	en-US learner AI respects selected variety.
AI	AIA-06	P9/P13	AI/application + QA/security	Free-form AI practice is ephemeral/practice-only; no direct mastery.
AI	AIA-07	P9/P13	AI/application + QA/security	Template-constrained generated item validates before bounded evidence.
AI	AIA-08	P9/P13	AI/application + QA/security	Invalid writing-feedback JSON falls back safely; original retained; no corrupt mastery.
AI	AIA-09	P8/P13	AI/application + QA/security	Prompt-injected publish/role action denied and audited.
AI	AIA-10	P8/P13	AI/application + QA/security	AI Auto may create draft/media job only; diff/audit/quota present; publish human.
AI	AIA-11	P8/P9/P13	AI/application + QA/security	Removed model falls back or becomes unavailable; core unaffected.
AI	AIA-12	P5/P9/P13	AI/application + QA/security	AI role-play mic denial/low confidence -> speech_unavailable; non-mic route; no penalty.
AI	AIA-13	P9/P13	AI/application + QA/security	Clearing AI conversation applies retention/deletion lifecycle.
AI	AIA-14	P8/P9/P13	AI/application + QA/security	Paid provider blocked pre-call when paid inference disabled; zero cost logged.
AI	AIA-15	P8/P9/P13	AI/application + QA/security	Prompt/model change requires bilingual/security eval before learner-default promotion.
AI	AIA-16	P5/P8/P13	AI/application + QA/security	AI-generated required image enters normal media pipeline and cannot publish before approved asset_version.

Trace rule: every MUST row has an approved source -> milestone -> owner -> verification mapping in this appendix. P0 closes only after these mappings are materialized as concrete backlog/test/ADR links in /docs/baseline/requirements-trace.csv and reviewed; P1 must not start earlier.

Canonical external references

- OpenAPI Specification 3.1.2
- CEFR Companion Volume
- Council of Europe - CEFR
- English Profile
- Čeština pro cizince

OpenAPI here means the OpenAPI Specification for describing HTTP APIs; it is unrelated to OpenAI as an AI provider. OpenAI remains only a possible future provider behind the provider-agnostic AI adapter and is not required by the v1 baseline.

Approval record

Поле	Значення
Baseline status	OWNER-LOCKED IMPLEMENTATION BASELINE / P0 AUTHORIZED
Approval date	10.09.2026
Approver	Product owner
Candidate version	v1.4
Accepted exceptions	None. Targeted closure audit: BLOCKER 0, IMPORTANT 0; final MINOR editorial residues corrected before owner lock.
Execution authority	Lingvessa_Implementation_Plan v1.4; P0 must be closed before P1.

Canonical machine-readable source: /packages/activity-schemas/mastery-registry.json. CI MUST validate every evidence_map against the registry and eligible skill-facet pair matrix; copied tables in specifications are checksum-checked mirrors.